Boundary CFT_3-on-S^3 observables on the truncated GeoVac spectral triple: partition function and wedge KMS entropy

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Abstract

We exhibit three complementary boundary-side quantities, framed by standard CFT_3-on-S^3 calculations, on the truncated GeoVac spectral triple, each obtained by a structurally distinct operation — (A) from the spectra of the truncated scalar and Dirac operators, (B) from the wedge KMS state $\rho_W = e^{-K_\alpha^W}/Z$ built from a posited wedge generator, (C) by comparing that generator with a continuum one: *(A)* the universal F-theorem coefficient F_{S^3}, extracted via the spectral-zeta derivative $-\frac{1}{2}\zeta'_\Delta(0)$, reproduces the Klebanov–Pufu–Safdi (2011) continuum value bit-exactly for both the conformally coupled scalar ($F_s = \frac{\log 2}{8} - \frac{3\zeta(3)}{16\pi^2}$) and the free Camporesi–Higuchi Dirac ($F_D = \frac{\log 2}{4} + \frac{3\zeta(3)}{8\pi^2}$); *(B)* the wedge KMS entropy has the exact asymptotic $S(\rho_W) = 2\log n_{\max} + (\coth 1 - \log(2\sinh 1)) + O(1/n_{\max})$, whose slope 2 is the exponent of the $\sim n_{\max}^2$ degeneracy of the lowest shell of the posited generator K_α^W (the slope is independent of the inverse temperature; the constant is the unit-β value, and ρ_W is not the Bisognano–Wichmann-normalised state, which would be $e^{-\pi K_\alpha^W}/Z$ since $K_\alpha^W = 2J_z$; an earlier reading of this 2 as the dimension of the entangling surface is withdrawn: for a CFT_3 that surface is a circle); *(C)* the framework's wedge generator $K_\alpha^W = J_{\mathrm{polar}}$ (a posited rotation generator, not derived from a state) shares a discrete integer spectrum and closure at rotation angle 2π (its adjoint flow closes at $t = \pi$, since $K_\alpha^W = 2J_z$) with the *Euclidean* rotation generator whose Wick rotation is the continuum Casini–Huerta–Myers modular Hamiltonian of a hemispherical cap of the spatial S^2 (the latter itself generates a boost-like flow with continuous spectrum); this is a structural *analogy* with the Euclidean generator only, and the identification of K_α^W as the modular Hamiltonian of a wedge-restricted state is **[OPEN]** (2026-09-28; see [11]). The scalar/Dirac partition functions decompose orthogonally under the master Mellin engine of [2]: $F_D + 2F_s = \log(2)/2$ isolates the M2 (Seeley–DeWitt) sector, $F_D - 2F_s = 3\zeta(3)/(4\pi^2)$ isolates the M3 (half-integer Hurwitz / odd-zeta) sector. This is the first verification of the master Mellin engine on non-GeoVac-internal physics observables. The F-theorem values themselves are due to [22] and are not new; the present paper's contributions are (i) their bit-exact reproduction from the discrete truncated spectral triple, (ii) the orthogonal M2/M3 dual-basis decomposition (Theorem 3.3), and (iii) the exact large-$n_{\max}$ asymptotic of the wedge-KMS entropy, a lowest-shell degeneracy count. The CHM comparison (C) is recorded as an analogy only; an earlier version claimed a finite-cutoff identification at GH-convergence rate, which is withdrawn (2026-09-28): K_α^W is posited, and a GH distance between spectral triples does not bound the convergence of one operator to another. The bulk-side identifications (Ryu–Takayanagi minimum surfaces, JLMS bulk-modular reconstruction, AdS_4 / H^4 dual geometry) are blocked at the framework's existing infrastructure: zero AdS / H^4 machinery, and the Wick-rotation $S^3 \to H^3$ supplies no natural discrete subgroup [5].

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1 Introduction

The GeoVac spectral triple [1, 6] is a finite-cutoff operator-system truncation of the Camporesi–Higuchi spectral triple [19] on the round S^3. Its modular structure on the hemispheric wedge has been developed across Papers 42 and 44–49 [11–17] (the Krein-carrier extension, formerly Paper 43, is now Paper 42 § 11): with the posited geometric BW-α generator $K_\alpha^W = J_{\text{polar}}$ (integer spectrum) and the Tomita–Takesaki BW-γ modular operator of the Gibbs state built from it, the flow closes, $\sigma_{2\pi}(\mathcal{O}) = \mathcal{O}$, bit-exactly — automatically, for any integer-spectrum generator, so this is not a finite-cutoff identification of the four-witness Wick-rotation theorem (withdrawn 2026-09-28; deriving the generator from a Camporesi–Higuchi wedge state is open, Paper 42). The Lorentzian *propinquity* extension to signature $(3,1)$ is descoped: Paper 45 [13] proves a degeneracy theorem (the natural Lorentzian-propinquity seminorm vanishes identically, closing that route) and the strong-form construction [14] is correspondingly descoped on the natural substrate, so a Lorentzian quantum-metric convergence remains an open target. What survives is the norm-resolvent non-compact-carrier closure [15] at the spectral level and Paper 45's signature-agnostic product-carrier rate. The Krein–Mondino–Sämann bridge [16] and the OSLPLS strong-form bridge [17] provide the synthetic-Lorentzian geometric interpretation.

The present paper turns from the math.OA structural content of the modular machinery to its *physics-side observables*. We ask whether the truncated spectral triple — its spectra, and the wedge KMS state ρ_W built on it — evaluated on boundary observables corresponding to standard CFT_3-on-S^3 calculations, reproduces the known continuum values, and whether the resulting structural pattern exhibits any genuinely new content beyond a numerical match.

Headline results. We establish three complementary boundary-side observables and one structural decomposition theorem:

1. **Partition function match** (Sec. 3): the universal F-theorem coefficient extracted from the framework's spectral-zeta machinery agrees *bit-exactly* with the Klebanov–Pufu–Safdi (2011) [22] continuum values for both the conformally coupled scalar and the free Camporesi–Higuchi Dirac on the round S^3.

2. **Master Mellin engine decomposition** (Theorem 3.3): the scalar and Dirac partition functions decompose into linear combinations of $\log 2$ and $\zeta(3)/\pi^2$ that project orthogonally onto the M2 and M3 sectors of the master Mellin engine [2]. This is the first verification of the master Mellin engine on non-GeoVac-internal physics observables.

3. **Wedge KMS entropy structural classification** (Sec. 4): the von Neumann entropy of ρ_W scales as $S(\rho_W) \sim 2\log n_{\max}$; the slope 2 is the exponent of the $n_{\max}(n_{\max}+1) \sim n_{\max}^2$ degeneracy of the lowest shell of the posited generator K_α^W **[MEASURED]** (degeneracy count, checked against direct diagonalisation) — a property of the truncation, not a continuum geometric dimension. The F-coefficient does *not* appear in $S(\rho_W)$, demonstrating that the framework's spectral-zeta side and wedge KMS state side encode *different* quantities on the same truncated triple.

4. **Casini–Huerta–Myers analogy** (Sec. 5): the framework's posited wedge generator K_α^W shares integer spectrum and 2π-periodic flow with the Euclidean rotation generator about the entangling circle, whose Wick rotation is the continuum CHM modular Hamiltonian of a hemispherical cap H_+ of the spatial S^2, $K_{\text{cont}} = 2\pi \int_{H_+} \xi \cdot T^{00}\, dA$ [30]; K_{cont} itself generates a boost-like flow with continuous spectrum, so the analogy is with the Euclidean generator only. **[OPEN]**: identifying K_α^W as a modular Hamiltonian (deriving it from a wedge-restricted state) and any operator-level convergence $K_\alpha^W \to K_{\text{cont}}$; the Paper 38 GH rate concerns the spectral triples, not this operator.

Bulk-side scope. Bulk-side identifications — Ryu–Takayanagi minimum-surface entanglement entropy [33], the JLMS [34] bulk-modular reconstruction, and AdS_4 / H^4 dual geometry — are blocked at the framework's existing infrastructure: zero AdS_4 / H^4 machinery exists, and a previous attempt at the natural Wick continuation $S^3 \to H^3$ [5] found no discrete subgroup $\Gamma \subset SO(3,1)$ selectable from framework invariants. We document this scope statement honestly in Sec. 6, with no claim that the bulk side is reachable from the framework's current axiom set.

2 Setup

2.1 The GeoVac spectral triple on S^3

The Fock projection [1, 18] maps the single-particle hydrogen Hamiltonian conformally to the Laplace–Beltrami operator on S^3 via stereographic projection in momentum space. The framework's spectra on the unit S^3 are exact in $\mathbb{Q}$:

- **Scalar Laplacian.** Eigenvalues $\lambda_n^2 = n(n+2)$ with multiplicity $d_n = (n+1)^2$ for $n = 0, 1, 2, \ldots$. After the conformal mass shift $\xi R_{\text{scalar}} = \frac{1}{8} \cdot 6 = \frac{3}{4}$ for the conformally coupled scalar, the spectrum becomes $(n+\frac{1}{2})(n+\frac{3}{2})$ with the same multiplicity.
- **Camporesi–Higuchi Dirac.** Eigenvalues $|\lambda_n| = n + \frac{3}{2}$ with multiplicity $g_n^{\text{Dirac}} = 2(n+1)(n+2)$ for the single (3d) Dirac fermion (the standard S^3 massless Dirac, whose F_D is reproduced below; in odd dimension there is no Weyl projection), $4(n+1)(n+2)$ for the doubled four-component Dirac. Both carry the same Hurwitz-zeta structure under analytic continuation [3, 4].

The truncation at level $n_{\max}$ retains all $n \le n_{\max}$ shells and defines the operator system $\mathcal{O}_{n_{\max}} = P_{n_{\max}} \mathcal{A} P_{n_{\max}}$ in the Connes–van Suijlekom framework [20]. Convergence $\mathcal{O}_{n_{\max}} \to \mathcal{O}_{S^3}$ in van Suijlekom's state-space Gromov–Hausdorff distance holds at rate $\gamma_{n_{\max}} \sim (4/\pi)\log(n_{\max})/n_{\max}$ [8]; the rate constant $4/\pi$ is stated in the corpus's dual-Coxeter normalisation ($\text{Cas}(\text{ad}) = h^\vee$, under which the $SU(2)$ geodesic distance is the rotation angle) and is convention-dependent, not canonical: on the unit S^3 the same bound reads $2/\pi = \text{Vol}(S^2)/\text{Vol}(S^3)$, the volume quotient that carries its M1 content in the master Mellin engine [2] (the earlier "Hopf-base" label is a misnomer).

2.2 Wedge KMS state

The hemispheric wedge W corresponds, in the truncated full-Dirac basis, to the half-space with $m_j > 0$ along the wedge-fixing axis. Per [11] Def 5.1, the posited BW-α wedge generator (its identification as a modular Hamiltonian is open) is

$$K_\alpha^W = J_{\text{polar}}$$

where J_{polar} is the rotation generator preserving the wedge, with integer eigenvalues $2m_j \in \{1, 3, 5, \ldots, 2n_{\max}-1\}$ and multiplicity $g(2k+1) = (n_{\max}-k)(n_{\max}-k+1)$ for $k = 0, \ldots, n_{\max}-1$. The wedge state used throughout is the unit-β Gibbs state of K_α^W (not the Bisognano–Wichmann-normalised state; see Remark 5.1)

$$\rho_W = \frac{e^{-K_\alpha^W}}{Z}, \qquad Z = \text{Tr}_{\mathcal{H}_W}\left(e^{-K_\alpha^W}\right).$$

This is the substrate for all three boundary-side observables that follow.

3 Partition function: bit-exact match to KPS

3.1 Continuum reference

For the round unit S^3, the free energies of free CFTs computed via Hurwitz-zeta regularization are [22]:

$$F_s^{\mathrm{KPS}} = \frac{\log 2}{8} - \frac{3\zeta(3)}{16\pi^2} \approx 0.063807, \tag{1}$$

$$F_D^{\mathrm{KPS}} = \frac{\log 2}{4} + \frac{3\zeta(3)}{8\pi^2} \approx 0.218960, \tag{2}$$

for the conformally coupled scalar and the free massless Dirac fermion (no Weyl projection in odd dimension), respectively. Both values are expressed in terms of the functional determinant identities $F = -\frac{1}{2}\zeta'_\Delta(0)$ (scalar) and $F = \zeta'_{|D|}(0)$ (Dirac), with the analytic continuation of the divergent spectral sum supplied by the standard zeta-function regularization. The transcendental signature involves only $\log 2$, $\zeta(3)$, and π^2.

3.2 Framework's Hurwitz machinery

The framework's spectral zeta on the truncated S^3 admits a closed-form Hurwitz expansion. For the Dirac case, the Dirichlet series can be written exactly via the half-integer Hurwitz identity $\zeta_H(s, 3/2) = (2^s - 1)\zeta_R(s) - 2^s$:

$$D_{\mathrm{Dirac}}(s) = \sum_{n=0}^{\infty} \frac{g_n^{\mathrm{Dirac}}}{|\lambda_n|^s} = 2\big(2^{s-2} - 1\big)\zeta_R(s-2) - \tfrac{1}{2}\big(2^s - 1\big)\zeta_R(s). \tag{3}$$

This identity is encoded in the production module `geovac/qed_two_loop.py` and is verified symbolically across integer s.

For the conformally coupled scalar, the spectral zeta admits the Hurwitz-asymptotic expansion

$$\zeta_\Delta(s) = \sum_{n=0}^{\infty} \frac{(n+1)^2}{[(n+1/2)(n+3/2)]^s} = \sum_{k\geq 0} g_k(s)\, \zeta_R(2s + 2k - 2), \tag{4}$$

with $g_k(s) = \binom{s+k-1}{k}/4^k$ the binomial-like coefficient.

3.3 Bit-exact match: scalar

Differentiating (4) at $s = 0$ and using $g_k(0) = \delta_{k,0}$, $g'_k(0) = 1/(k \cdot 4^k)$ for $k \geq 1$, plus the standard identities $\zeta'_R(-2) = -\zeta(3)/(4\pi^2)$, $\zeta_R(0) = -\frac{1}{2}$, $\zeta_R(2k-2)$ expressible in $\pi^{2k-2} \cdot \mathbb{Q}$ for $k \geq 2$:

$$\zeta'_\Delta(0) = 2\zeta'_R(-2) + \sum_{k\geq 1} \frac{\zeta_R(2k-2)}{k \cdot 4^k}. \tag{5}$$

Theorem 3.1 (Bit-exact scalar partition function)**.** *The framework's scalar zeta-derivative satisfies*

$$F_s := -\tfrac{1}{2}\,\zeta'_\Delta(0) = \frac{\log 2}{8} - \frac{3\zeta(3)}{16\pi^2} = F_s^{KPS}, \tag{6}$$

in bit-exact agreement with (1).

Proof sketch. The asymptotic Hurwitz expansion (5) converges geometrically in k at rate $1/4$ per term; truncation at $k_{\max} = 100$ gives 61+ matching digits at dps = 200 when compared to $-2F_s^{\mathrm{KPS}}$ via `mpmath`. PSLQ at coefficient bound 10^6 and tolerance 10^{-40} finds the integer relation $[8, 2, -3]$:

$$8 \cdot \zeta'_\Delta(0) + 2\log 2 - 3 \cdot \frac{\zeta(3)}{\pi^2} = 0,$$

which is exactly the claimed identity. See driver `debug/ads_track_a_scalar_partition_function.py` and JSON `debug/data/ads_track_a_scalar_partition_function.json` for full computation. □

3.4 Bit-exact match: Dirac

Differentiating (3) symbolically at $s = 0$, using the same Riemann-zeta identities at $s = -2, 0$:

$$D'_{\text{Dirac}}(0) = \tfrac{1}{2}\log 2 \cdot \zeta_R(-2) + \left(-\tfrac{3}{2}\right)\zeta'_R(-2) - \tfrac{1}{2}\left[\log 2 \cdot \zeta_R(0) + 0\right] \tag{7}$$

$$= 0 - \tfrac{3}{2} \cdot \left(-\tfrac{\zeta(3)}{4\pi^2}\right) - \tfrac{1}{2} \cdot \left(-\tfrac{\log 2}{2}\right) = \frac{3\zeta(3)}{8\pi^2} + \frac{\log 2}{4}. \tag{8}$$

Theorem 3.2 (Bit-exact Dirac partition function)**.** *The framework's Dirac zeta-derivative satisfies, as an exact symbolic identity,*

$$F_D := D'_{\text{Dirac}}(0) = \frac{\log 2}{4} + \frac{3\zeta(3)}{8\pi^2} = F_D^{KPS}, \tag{9}$$

in bit-exact agreement with (2).

Proof. Direct symbolic computation; `sympy.simplify(D'(0) - F_D)` returns 0. See driver `debug/ads_track_a_dirac_partition_function.py`. □

3.5 Master Mellin engine decomposition

The master Mellin engine of [2] §III.7 classifies the transcendental signatures of GeoVac observables via the three sub-mechanisms

$$\mathcal{M}[\mathrm{Tr}(D^k \cdot e^{-tD^2})], \qquad k \in \{0, 1, 2\},$$

labeled M1 (Hopf-base measure, $\sqrt{\pi} \cdot \mathbb{Q}$ ring at $k = 0$), M2 (Seeley–DeWitt heat-kernel, $\sqrt{\pi} \cdot \mathbb{Q} \oplus \pi^2 \cdot \mathbb{Q}$ ring at $k = 2$), and M3 (half-integer Hurwitz / vertex-parity, $\zeta(3)/\pi^2$ and similar odd-zeta-over-pi ratios at $k = 1$).

Theorems 3.1 and 3.2 exhibit F_s and F_D as linear combinations of $\log 2$ (an M2 transcendental, present via $\zeta'_R(0) = -\frac{1}{2}\log(2\pi)$ in the underlying heat-kernel asymptotic) and $\zeta(3)/\pi^2$ (an M3 transcendental, present via $\zeta'_R(-2) = -\zeta(3)/(4\pi^2)$ in the half-integer Hurwitz-zeta derivative).

Theorem 3.3 (Scalar/Dirac M2–M3 orthogonal decomposition)**.** *The free energies F_s and F_D project orthogonally onto the M2 and M3 axes of the master Mellin engine via the dual-basis combinations*

$$F_D + 2\,F_s = \frac{\log 2}{2} \qquad \textit{(isolates M2; all } \zeta(3) \textit{ cancels)}, \tag{10}$$

$$F_D - 2\,F_s = \frac{3\zeta(3)}{4\pi^2} \qquad \textit{(isolates M3; all } \log 2 \textit{ cancels)}. \tag{11}$$

Proof. Substituting (1) and (2):

$$F_D + 2F_s = \left(\tfrac{\log 2}{4} + \tfrac{3\zeta(3)}{8\pi^2}\right) + 2\left(\tfrac{\log 2}{8} - \tfrac{3\zeta(3)}{16\pi^2}\right) = \tfrac{\log 2}{4} + \tfrac{\log 2}{4} = \tfrac{\log 2}{2},$$

and similarly $F_D - 2F_s = 3\zeta(3)/(4\pi^2)$. The two combinations span a 2-dimensional space; the projection coefficients $(1, +2)$ and $(1, -2)$ act as a dual basis for the M2/M3 decomposition on the (scalar, Dirac) plane. □

Remark 3.4 (First verification on non-GeoVac-internal data)**.** Theorem 3.3 is the first verification of the master Mellin engine on observables that are *not* GeoVac-internal. Previous verifications — the case-exhaustion theorem of [6] §VIII, the 208/208 Prediction 1 panel of [7], the modular-residual closed form of Sprint MR-B[1] — all operated on framework-internal quantities. The KPS partition functions (1), (2) are independently-published continuum CFT_3 data, and the framework's spectral-zeta machinery reproduces them with the exact M2/M3 transcendental signature the master Mellin engine predicts. This extends the engine's reach from a structural classification of GeoVac-internal observables to the organization of independently-published CFT_3 data on the round S^3 at finite cutoff — a bit-exact *reproduction* of the KPS values, not an independent prediction of CFT physics.

4 Wedge KMS entropy: state-side observable

4.1 The framework's wedge KMS entropy

The von Neumann entropy of the wedge state $\rho_W = e^{-K_\alpha^W}/Z$ (unit inverse temperature for K_α^W; not the Bisognano–Wichmann normalisation, Remark 5.1) is computed directly by diagonalizing K_α^W and applying $S(\rho_W) = -\mathrm{Tr}[\rho_W \log \rho_W]$ (recomputed from the live BW construction, the slope-2 scaling, the $\coth 1 - \log(2\sinh 1)$ constant, and the area-law rejection, in `tests/test_paper50_wedge_kms.py`). Sprint BH-Phase0 (2026-05-22) [36] first reported a small-cutoff log-linear fit $S(\rho_W) \approx 1.94 \log n_{\max} + 0.56$ over $n_{\max} \in \{2, \ldots, 7\}$. A subsequent reanalysis (forcing-catalogue Door 2, 2026-06-01, `debug/door2_bw_entropy_reread_memo.md`) replaced this fit by the exact $n_{\max} \to \infty$ resummation of the closed-form entropy $S = \log Z + \langle K_\alpha^W \rangle$ (which reproduces the direct diagonalization to $\leq 4 \times 10^{-16}$). The leading $n_{\max}^2$ shell degeneracy $g(2k+1) = (n_{\max} - k)(n_{\max} - k + 1)$ forces the slope to be *exactly* 2, and the additive constant has the exact closed form

$$S(\rho_W) = 2 \log n_{\max} + \big(\coth 1 - \log(2 \sinh 1)\big) + O(1/n_{\max}), \qquad \coth 1 - \log(2\sinh 1) = 0.458448\ldots, \tag{12}$$

the constant verified by PSLQ to residual $< 10^{-80}$ (integer relation $[-1, 1, -1, -1]$ against $\{C_\infty, \coth 1, \log 2, \log \sinh 1\}$). The windowed least-squares slope climbs monotonically $1.94 \to 2.00005$ as the fit window moves to $n_{\max} \in [350, 2000]$; the original 1.94 is a small-$n_{\max}$ artifact of a sequence converging to exactly 2. The slope 2 is the exponent of the lowest-shell degeneracy $n_{\mathrm{eq}} = n_{\max}(n_{\max}+1)$ of the posited generator K_α^W (mechanism below) **[MEASURED]**; an earlier version read it as the dimension of the entangling surface, which is withdrawn (2026-09-28): for a CFT_3 the entangling surface of a hemispherical cap is a circle S^1, not a two-sphere. The additive constant lives in the Boltzmann-base ring (functions of e via $\coth 1$, $\sinh 1$, tied to the integer spacing $\Delta K_\alpha^W = 2$ at unit β); it carries no master Mellin engine transcendental, consistent with Prop. 4.1. A PSLQ search against the F-theorem ring $\{1, \log 2, \zeta(3)/\pi^2\}$ at coefficient ceiling 10^8 returns null: the wedge-entropy constant and the F-coefficient lie in disjoint rings, sharpening Prop. 4.1 from "F absent from $S(\rho_W)$" to "the state-side constant admits no integer relation with the spectral-zeta ring up to coefficient ceiling 10^8" (a strong numerical disjointness, not a proof of ring non-membership). The structural support is separate for the two terms: the leading $2 \log n_{\max}$ is the logarithm of an integer degeneracy, hence ring-free; the additive constant is *not* the logarithm of an integer, but a closed-form function of e alone ($\coth 1$, $\sinh 1$), i.e. in the Boltzmann-base ring by construction.

4.2 Mechanism: degeneracy log on the lowest K_α^W shell

The K_α^W spectrum is bounded above by $2n_{\max} - 1$ at finite cutoff. In ρ_W (unit β), the per-state weight at $K_\alpha^W = 1$ (the lowest shell) is e^{-1}/Z, while at $K_\alpha^W = 3$ it drops to e^{-3}/Z, a factor of

[1] Sprint/track designations are anchors into the project's internal chronicle (`CHANGELOG.md` in the repository).

$e^{-2} \approx 0.135$ suppression per shell. The partition function is therefore dominated by the lowest shell of multiplicity $n_{\mathrm{eq}} = n_{\max}(n_{\max}+1) \sim n_{\max}^2$:

$$Z \approx n_{\mathrm{eq}} \cdot e^{-1} \approx 0.368\, n_{\max}^2,$$

giving effective probability $p_{\mathrm{eq}} \approx 1/n_{\mathrm{eq}}$ per lowest-shell state, and hence

$$S(\rho_W) \approx -n_{\mathrm{eq}} \cdot p_{\mathrm{eq}} \log p_{\mathrm{eq}} = \log n_{\mathrm{eq}} \approx 2 \log n_{\max}.$$

The two asymptotic limits in s-deformation $\rho(s) = e^{-sK_\alpha^W}/Z$, $s \to 0^+$ (maximally mixed on $\mathcal{H}_W$, giving $S \to \log \dim \mathcal{H}_W$) and $s \to \infty$ (maximally mixed on the lowest shell, giving $S \to \log n_{\mathrm{eq}}$), are saturated bit-exactly [36].

4.3 Structural decomposition: F is in the spectral-zeta side, not the state side

Proposition 4.1 (F vs $S(\rho_W)$ live in different sectors)**.** *The truncated spectral triple admits two structurally distinct calculational operations that extract different quantities:*

- ***Spectral-zeta side:*** *the analytic continuation $-\frac{1}{2}\zeta'_\Delta(0)$ extracts the universal F-theorem coefficient F_s^{KPS} via Theorem 3.1; similarly for F_D^{KPS} via Theorem 3.2. The transcendental signature is in the master Mellin engine M2/M3 ring $\mathbb{Q} \cdot \log 2 \oplus \mathbb{Q} \cdot \zeta(3)/\pi^2$.*
- ***Wedge KMS state side:*** *the von Neumann entropy $S(\rho_W) = -\mathrm{Tr}[\rho_W \log \rho_W]$ extracts the lowest-shell degeneracy of the posited generator K_α^W, with leading scaling $2 \log n_{\max}$ (the degeneracy grows as $n_{\max}^2$). The leading term is ring-free (a logarithm of an integer).*

The two operations act on the same truncated triple (the first on its spectra, the second on the wedge state ρ_W built from it) but extract categorically different *quantities: the universal F-coefficient versus a lowest-shell degeneracy count. The F-coefficient does* not *appear in $S(\rho_W)$.*

This decomposition is consistent with the PSLQ-disjointness finding of Sprint TD Track 5 (2026-05-09) [37] that GeoVac entropies generally live outside the master Mellin engine ring, while spectral-side observables generally live inside it.

4.4 Three blockers for literal Ryu–Takayanagi on the framework's graph

The continuum Ryu–Takayanagi formula [33] identifies the boundary entanglement entropy of a region A with the area of the bulk minimum surface anchored at ∂A, weighted by $1/(4G_N)$. A literal discrete analog on the framework's graph would require: (i) a well-defined notion of graph minimum-cut between spatial sub-regions; (ii) a perfect-tensor structure at graph vertices ensuring isometric bulk-boundary encoding; (iii) a bulk dual geometry.

Proposition 4.2 (Three blockers for literal RT on the framework)**.** *The framework's discrete S^3 graph at finite $n_{\max}$ exhibits three independent structural obstructions to a literal RT formula:*

1. *The Fock graph at level $n_{\max}$ has $\beta_0 = n_{\max}$ connected components (one per ℓ-sector), so the naive graph-cut between spatial sub-regions is identically zero or trivially confined to a single ℓ-sector.*
2. *The Wigner 3j symbols at graph vertices are Clebsch–Gordan projections rather than isometries; the HaPPY tensor-network RT proof [35] requires perfect tensors at network nodes.*
3. *No bulk-dual construction exists in the framework's infrastructure. The natural candidate — Wick continuation $S^3 \to H^3$ — was closed by Sprint RH-B (2026-04-17) [5] as a clean structural dead end: no discrete subgroup $\Gamma \subset SO(3,1)$ is selectable from framework invariants.*

These three blockers are not in tension with the boundary-side results of Sec. 3 or Sec. 5: the F-coefficients of Sec. 3 are computed from the operator spectra (not from the wedge state), the wedge entropy is a lowest-shell degeneracy count, and the CHM comparison is an analogy; none of these requires, and the framework cannot independently construct, a bulk dual against which to compare via RT.

5 Casini–Huerta–Myers analogy

5.1 Continuum CHM hemisphere modular Hamiltonian

For a CFT_3 the Lorentzian spatial slice is a two-sphere, and the Casini–Huerta–Myers construction [30] (valid for any CFT, not only free ones) gives the modular Hamiltonian of the vacuum restricted to a hemispherical cap $H_+ \subset S^2$ as

$$K_{\mathrm{cont}} = 2\pi \int_{H_+} \xi(x) \cdot T^{00}(x)\, dA(x), \tag{13}$$

where ξ is the conformal Killing vector field preserving the causal development of the cap (vanishing on the entangling surface $\partial H_+ = S^1$, the boundary circle of the cap), and T^{00} is the local energy density. The CHM map sends the cap's causal development conformally to $\mathbb{R} \times H^2$, carrying the restricted vacuum to a thermal state at $\beta = 2\pi$ (in units of the curvature radius) [30, 31]. In the Lorentzian theory ξ generates a *boost-like* conformal flow. That K_{cont} has continuous spectrum and that the cap algebra is of type III are not statements of CHM or of Bisognano–Wichmann; they are standard facts of algebraic quantum field theory (under standard conditions local algebras are type III_1 factors [32], and for a type III_1 factor Connes' invariant $S(\mathcal{M})$, the intersection of the modular spectra, is the whole half-line $[0, \infty)$). The Bisognano–Wichmann 2π is a KMS parameter, not a periodicity of the real-time flow. Only the *Euclidean* continuation of this flow has discrete spectrum and a periodic flow: the Euclidean section is (conformally) S^3, and the continued flow is the rotation of S^3 that fixes a great circle (the Euclidean image of the entangling circle), closing at angle 2π; the KMS $\beta = 2\pi$ is exactly that Euclidean periodicity. The comparison below is with this Euclidean rotation generator, not with K_{cont} itself.

5.2 Framework wedge generator

The framework's posited wedge generator (Paper 42 Def 5.1) is

$$K_\alpha^W = J_{\mathrm{polar}},$$

the rotation generator of the wedge-preserving $SO(2)$ on the truncated Camporesi–Higuchi spectral triple. Its spectrum is

$$\mathrm{spec}(K_\alpha^W) = \{2m_j : m_j = 1/2, 3/2, \ldots, (2n_{\max} - 1)/2\} = \{1, 3, 5, \ldots, 2n_{\max} - 1\},$$

integer-valued ($K_\alpha^W = 2J_z$), with multiplicity $g(2k+1) = (n_{\max} - k)(n_{\max} - k + 1)$. The posited modular flow $\sigma_t = \mathrm{Ad}\, e^{itK_\alpha^W}$ satisfies $\sigma_{2\pi}(\mathcal{O}) = \mathcal{O}$ bit-exactly at every tested $n_{\max}$ [11]. The odd-integer spectrum, the multiplicities, and the 2π closure are verified from the live BW construction in `tests/test_paper50_wedge_kms.py`.

Remark 5.1 (Normalisation of K_α^W and of ρ_W)**.** Since $\mathrm{spec}(K_\alpha^W) = \{2m_j\}$, $K_\alpha^W = 2J_z$, where J_z generates rotations by angle θ about the wedge axis ($e^{-i\theta J_z}$). Three consequences. (i) The adjoint flow $\mathrm{Ad}\, e^{itK_\alpha^W}$ already closes at $t = \pi$: differences of odd integers are even. The 2π closure above holds but is twice the minimal period; the unitary $e^{itK_\alpha^W}$ itself first returns to 1 at $t = 2\pi$ (it is -1 at $t = \pi$, the spinor sign). In the rotation-angle parameter $\theta = 2t$ the adjoint

flow closes at $\theta = 2\pi$, which is the Euclidean periodicity matched in Observation 5.2. (ii) A state normalised as in Bisognano–Wichmann, with KMS parameter 2π in the rotation angle, would be $e^{-2\pi J_z}/Z = e^{-\pi K_\alpha^W}/Z$, not $\rho_W = e^{-K_\alpha^W}/Z$; ρ_W is the unit-β Gibbs state of K_α^W (inverse temperature 2 in rotation-angle units). (iii) For $\rho_b = e^{-bK_\alpha^W}/Z$ one has $Z \simeq n_{\max}^2/(2\sinh b)$ and $\langle K_\alpha^W \rangle \to \coth b$, so $S(\rho_b) = 2\log n_{\max} + b\coth b - \log(2\sinh b) + O(1/n_{\max})$: the slope-2 headline is independent of b, while the constant $\coth 1 - \log(2\sinh 1)$ of (12) is the $b = 1$ value and is *not* a Bisognano–Wichmann-normalised quantity (at $b = \pi$ the constant would be $\pi\coth\pi - \log(2\sinh\pi) = 0.0136\ldots$).

Observation 5.2 (CHM analogy at finite cutoff)**.** ***[SYMBOLIC]*** *(the framework-side items) /* ***[OPEN]*** *(the identification). The framework's posited wedge generator K_α^W at finite $n_{\max}$ is a structural analog of the* Euclidean *rotation generator whose Wick rotation is the continuum CHM modular Hamiltonian K_{cont} of (13) — not of K_{cont} itself, which has continuous spectrum:*

- *K_α^W has odd-integer spectrum; the Euclidean rotation generator, in the same $2J$ normalisation, has discrete (integer) spectrum (continuum statement, prior art).*
- *Each generates a flow closing at rotation angle 2π (for $K_\alpha^W = 2J_z$ the adjoint flow closes at $t = \pi$; Remark 5.1); for the continuum this is the Euclidean periodicity that the Lorentzian modular flow of K_{cont} exhibits only as its KMS parameter $\beta = 2\pi$. The state ρ_W used in Sec. 4 is not normalised to that β.*
- *The framework's odd-integer spectrum $\{1, 3, \ldots, 2n_{\max} - 1\}$ corresponds to the half-integer Casimir grading of the wedge-preserving Killing field on the (two-component, 3d) Dirac spinor sector.*

The integer spectrum and the 2π period are automatic for any integer-spectrum rotation generator [11], so they do not single out a modular Hamiltonian. What is not *established: that K_α^W is the modular Hamiltonian of a wedge-restricted Camporesi–Higuchi state (the open falsifier of [11]), and any convergence $K_\alpha^W \to K_{\text{cont}}$ as operators. The state-space Gromov–Hausdorff convergence of the spectral triples ([8], rate $(4/\pi)\log n_{\max}/n_{\max}$) is a statement about the triples, not about this operator; an earlier version of this item used it as an operator convergence rate, which is withdrawn (2026-09-28).*

Basis of the analogy. The wedge-preserving Killing field of the round (Euclidean) S^3 — the Euclidean rotation generator, not the Lorentzian boost-like K_{cont} — rotates the spinor harmonics by their m_j eigenvalue, giving the integer $2m_j$ spectrum. The framework's $K_\alpha^W = J_{\text{polar}}$ acts on the projected wedge sub-space $P_W \mathcal{H}_{\text{GV}}^{n_{\max}}$ with the same $2m_j$ eigenvalues, restricted to the multiplicities allowed by the truncation.

6 Bulk-side scope

The three boundary-side observables (Sec. 3, Sec. 4, Sec. 5) reproduce a continuum CFT_3-on-S^3 value (A), give an exact finite-cutoff degeneracy asymptotic (B), or parallel a continuum object (C, an analogy with the identification open). The corresponding *bulk-side* identifications — AdS_4 gravity, JLMS bulk-modular reconstruction, RT minimum surfaces, holographic entanglement entropy — are blocked at the framework's existing infrastructure.

Proposition 6.1 (Bulk side blocked)**.** *None of the bulk-side identifications corresponding to Sec. 3–5 are reachable from the framework's current infrastructure:*

1. *No AdS_4 / H^4 machinery exists in the framework's codebase (`geovac/`); a search for AdS, hyperbolic-4-space, or holographic primitives returns no matches.*

2. *Sprint RH-B (2026-04-17) [5] closed the natural candidate — Wick continuation $S^3 \to H^3$ — as a clean structural dead end: the continuation supplies no natural discrete subgroup $\Gamma \subset SO(3,1)$ selectable from framework invariants.*

3. *Sprint L3e-P3 (2026-05-23) found that Paper 38's $4/\pi$ rate does* not *transport to the non-compact Coulomb setting, suggesting that any discrete-to-bulk convergence statement would require a non-compact rate analog that is itself an open NCG-math problem.*

The structural reading of Prop. 6.1 is that the bulk-side identification falls into Class 1 (calibration-external) of the framework's external-input partition [38]: the framework determines the boundary-side observables structurally but does not autonomously generate the bulk-dual geometry. Importing AdS_4 infrastructure would be a major structural extension, beyond sprint scale and outside the present paper's scope.

7 Extension to S^5: structural refinement of the dual-basis decomposition

The bit-exact match of Sec. 3 extends to round S^5 via the same Hurwitz-zeta machinery. The Camporesi–Higuchi spectra on S^5 are spectrally analogous to S^3: the conformally coupled scalar has eigenvalues $(n+3/2)(n+5/2)$ with multiplicity $(2n+4)(n+1)(n+2)(n+3)/24$ for $n = 0, 1, 2, \ldots$, and the Weyl Camporesi–Higuchi Dirac has $|\lambda_n| = n + 5/2$ with multiplicity $(n+1)(n+2)(n+3)(n+4)/12$. The framework's spectral-zeta machinery on these spectra produces partition function values in a closed-form ring extending [22]'s S^3 ring by one $\zeta(5)/\pi^4$ generator.

7.1 Bit-exact closed forms on S^5

Theorem 7.1 (Conformally coupled scalar on S^5)**.** *The framework's scalar zeta-derivative on round unit S^5 satisfies*

$$F_s^{S^5} := -\tfrac{1}{2}\,\zeta'_{\Delta_{conf}}(0)^{S^5} = -\frac{\log 2}{128} - \frac{\zeta(3)}{128\pi^2} + \frac{15\,\zeta(5)}{256\pi^4} \approx -5.74 \times 10^{-3}. \tag{14}$$

Proof sketch. The Hurwitz expansion

$$\zeta^{\mathrm{conf}}_{S^5}(s) = \frac{1}{12}\sum_{k\geq 0} g_k(s)[\zeta_R(2s+2k-4) - \zeta_R(2s+2k-2)] \tag{15}$$

with $g_k(s) = \binom{s+k-1}{k}/4^k$ follows from the re-indexed multiplicity structure $v^2(v^2-1)/12$ at $v = n+2$. At $s=0$, only $g_0(0) = 1$ survives, giving $\zeta^{\mathrm{conf}}_{S^5}(0) = 0$; the derivative

$$\zeta'_{S^5}(0) = \frac{1}{6}[\zeta'_R(-4) - \zeta'_R(-2)] + \frac{1}{12}\sum_{k\geq 1}\frac{\zeta_R(2k-4) - \zeta_R(2k-2)}{k\cdot 4^k} \tag{16}$$

converges geometrically in k. PSLQ at 200 dps and $k_{\max} = 100$ finds the integer relation $[32, -2, -2, 15]$ (applied to $4\,\zeta'_{S^5}(0)$) bit-exactly identifying

$$\zeta'_{S^5}(0) = \frac{\log 2}{64} + \frac{\zeta(3)}{64\pi^2} - \frac{15\,\zeta(5)}{128\pi^4}. \tag{17}$$

The standard identities $\zeta'_R(-2) = -\zeta(3)/(4\pi^2)$ and $\zeta'_R(-4) = +3\zeta(5)/(4\pi^4)$ deliver the $\zeta(5)/\pi^4$ contribution; the $\log 2$ piece arises from $\zeta'_R(0) = -\frac{1}{2}\log(2\pi)$ via the Hurwitz analytic-continuation tail. □

Theorem 7.2 (Weyl Camporesi–Higuchi Dirac on S^5)**.** *The framework's Weyl Camporesi–Higuchi Dirac zeta-derivative on round unit S^5 satisfies, as an exact symbolic identity,*

$$D'_{Dirac}(0)^{S^5} = -\frac{3\log 2}{128} - \frac{5\,\zeta(3)}{128\pi^2} - \frac{15\,\zeta(5)}{256\pi^4} \approx -0.02163. \tag{18}$$

Proof sketch. The Dirichlet series

$$D_{\mathrm{Dirac}}(s)^{S^5} = \frac{1}{12}\left[\zeta_H(s-4,5/2) - \tfrac{5}{2}\zeta_H(s-2,5/2) + \tfrac{9}{16}\zeta_H(s,5/2)\right] \tag{19}$$

follows from re-indexing $u = n + 5/2$ in the multiplicity $(n+1)(n+2)(n+3)(n+4)/12 = (u^2 - 9/4)(u^2 - 1/4)/12$. The half-integer Hurwitz function $\zeta_H(s, 5/2)$ has closed form $(2^s - 1)\zeta_R(s) - 2^s - (2/3)^s$. Differentiating each shift symbolically at $s = 0$ and combining via the multiplicity coefficients $(1, -5/2, 9/16)$ delivers the claimed result. □

7.2 Log 3 cancellation theorem

Theorem 7.3 ($\log 3$ cancellation on S^5)**.** *The $(2/3)^s$ terms in $\zeta_H(s-a, 5/2)$ for $a \in \{0, 2, 4\}$ contribute, at $s = 0$, factors of $\log(2/3) = \log 2 - \log 3$ scaled by $(2/3)^{-a} = (3/2)^a$. The coefficient of $\log 3$ in $D'_{Dirac}(0)^{S^5}$ evaluates to*

$$\frac{1}{12}\left[\frac{81}{16} - \frac{5}{2}\cdot\frac{9}{4} + \frac{9}{16}\cdot 1\right] = \frac{1}{12}\left[\frac{81 - 90 + 9}{16}\right] = 0 \tag{20}$$

exactly.

Remark 7.4. The structural combination $(1, -5/2, 9/16)$ encoded in the multiplicity polynomial $(u^2 - 9/4)(u^2 - 1/4) = u^4 - (5/2)u^2 + 9/16$ aligns the three shift operators $\{\zeta_H(s, 5/2), \zeta_H(s - 2, 5/2), \zeta_H(s - 4, 5/2)\}$ such that all $\log 3$ contributions cancel exactly. The framework's spectral content on S^5 stays in the M-engine ring $\{\log 2, \zeta(3)/\pi^2, \zeta(5)/\pi^4\}$, even though the naive Hurwitz expansion suggests a contribution from $\log 3$. The same mechanism extends the M2/M3 ring statement from S^3 to S^5.

7.3 Dual-basis projection: structural-specificity to S^3

Theorem 3.3 of Sec. 3 establishes that on S^3, the (scalar, Dirac) pair admits a dual-basis projection onto the M2/M3 axes: $F_D + 2F_s = \log(2)/2$ (M2 only), $F_D - 2F_s = 3\zeta(3)/(4\pi^2)$ (M3 only). On S^5, the analogous statement *fails*.

Proposition 7.5 (Dual-basis non-extension to S^5)**.** *The Paper 50 Theorem 3.3 dual-basis projection does not extend to S^5. The M-engine ring on S^5 is three-dimensional ($\log 2$, $\zeta(3)/\pi^2$, $\zeta(5)/\pi^4$), but the (scalar, Dirac) plane is two-dimensional. The system $aF_s^{S^5} + bF_D^{S^5} = c \cdot$ (single M-axis) is over-determined: imposing isolation of one M-axis requires three linear conditions on two parameters (a, b), which has no nontrivial solution.*

Explicitly, attempting to isolate $\log 2$:

$$\zeta(3)/\pi^2 \text{ coefficient} = 0 \implies a + 5b = 0 \implies a = -5b, \tag{21}$$

$$\zeta(5)/\pi^4 \text{ coefficient} = 0 \implies 15a - 15b = 0 \implies a = b, \tag{22}$$

which are inconsistent unless $b = 0$.

Remark 7.6 (Continuation of the F-coefficient ladder through S^{11}; dual-basis non-extension)**.** Two statements must be distinguished here: the *single-species* in-ring closure of F (which continues to all odd d tested), and the *dual-basis projection* of Theorem 3.3 (which is special to S^3). The structural-specificity of Theorem 3.3 concerns only the second.

Single-species in-ring closure (continues through S^{11}). For each free species individually, $F = -\frac{1}{2}\zeta'_\Delta(0)$ on round S^d (d odd) lies in the ring

$$\mathcal{R}_d = \{\log 2\} \cup \{\zeta(2k+1)/\pi^{2k} : k = 1, \ldots, (d-1)/2\}$$

of dimension $(d-1)/2+1$ (top atom $\zeta(d)/\pi^{d-1}$). This is verified bit-exactly at S^3, S^5 (Theorems 7.1, 7.2), and — by direct high-precision computation (Door 1 graduation, 2026-06-03; 260 dps, frozen basis, reconstruction error $\lesssim 10^{-261}$, decoy-controlled) — at S^7, S^9, and S^{11}. The conformally coupled scalar on S^{11}, for instance, is

$$\begin{aligned}\zeta'_{\mathrm{sc},S^{11}}(0) = &-\tfrac{7}{131072}\log 2 - \tfrac{3897}{45875200}\tfrac{\zeta(3)}{\pi^2} - \tfrac{485}{8257536}\tfrac{\zeta(5)}{\pi^4} \\ &+ \tfrac{609}{1310720}\tfrac{\zeta(7)}{\pi^6} + \tfrac{425}{262144}\tfrac{\zeta(9)}{\pi^8} + \tfrac{1023}{524288}\tfrac{\zeta(11)}{\pi^{10}},\end{aligned} \tag{23}$$

and the free Weyl Camporesi–Higuchi Dirac value is symbolic-exact in the same ring. A decoy test assigns Catalan G, $\log 3$, and unscaled $\zeta(3)$ coefficient exactly zero, confirming a clean M3 signature.

Erratum. An earlier version of this remark reported S^7 as a structural non-match ("no PSLQ relation in the tested bases"). That was a *false negative* from an under-resolved search. The genuine S^7 scalar relation

$$\zeta'_{\mathrm{sc},S^7}(0) = -\tfrac{1}{512}\log 2 - \tfrac{41}{15360}\tfrac{\zeta(3)}{\pi^2} + \tfrac{5}{1024}\tfrac{\zeta(5)}{\pi^4} + \tfrac{63}{2048}\tfrac{\zeta(7)}{\pi^6}$$

has integer coefficients of order 10^4, but locking it requires the value to $\gtrsim 200$ digits; a 30-digit value run against maxcoeff $= 10^{16}$ sits below the resolution threshold and returns spurious non-detection. Verified independently at 200 dps (reconstruction error $\approx 2.4 \times 10^{-202}$) and reproduced by the Door 1 driver at 260 dps. The ladder therefore *generates*: every odd rung $S^3, \ldots, S^{11}$ has a rational-coefficient closed form in $\mathcal{R}_d$, with no transcendental outside $\mathcal{R}_d$.

Dual-basis projection (special to S^3, unaffected). Isolating a single M-axis using only the (scalar, Dirac) pair requires $\dim \mathcal{R}_d = 2$, which holds at S^3 alone. For $d \geq 5$ the projection is over-determined (Prop. 7.5); the related cross-species ladder recursion scalar $- m \cdot$ Dirac $= c_d \cdot \mathrm{Dirac}_{S^{d-2}}$ likewise closes at S^5 only, with the top-atom ratio $\mathrm{scalar}_{\mathrm{top}}/\mathrm{Dirac}_{\mathrm{top}} = 2^{-(d-5)/2}$ hitting the required value ($m = 1$) at $d = 5$ alone (Door 1 graduation). Thus the F-coefficients are in-ring at every odd d, but the two-species dual basis that *spans* the ring is special to S^3. Supplying the missing higher-spin free species — $(d-1)/2 - 1$ of them at S^d — to span $\mathcal{R}_d$ remains the natural continuation, and is the actual structural-specificity content of Theorem 3.3.

7.4 Implementation effort and continuation pattern

The S^5 extension was computed in 35 minutes of focused main-session work, continuing the discrete-tractability advantage documented across the math.OA arc. The `debug/ads_track_a_s5_{scalar, dirac}_partition_function.py` scripts implement the Hurwitz expansions and PSLQ verifications; `debug/ads_track_a_s5_extension_memo.md` documents the findings in full.

8 Synthesis and open questions

8.1 Unified picture

The truncated spectral triple, with the wedge KMS state $\rho_W = e^{-K_\alpha^W}/Z$ built on it, supports three structurally distinct calculational operations:

Operation	Continuum analog	Ring
$-\frac{1}{2}\zeta'_\Delta(0)$ (Theorem 3.1)	F-theorem coefficient	M2 + M3
$-\mathrm{Tr}[\rho_W \log \rho_W]$ (Prop. 4.1)	none claimed (lowest-shell degeneracy of K_α^W)	ring-free
$\mathrm{spec}(K_\alpha^W)$ (Observation 5.2)	Euclidean CHM rotation generator (analogy)	ring-free

The spectral triples on which all three are computed converge as $n_{\max} \to \infty$ in the van Suijlekom state-space Gromov–Hausdorff distance per [8] (with the signature-agnostic product-carrier rate of [13] on the temporal factor). For the first row the continuum value is reproduced; the second row is a finite-cutoff degeneracy count with no continuum identification claimed; the third row is an analogy with the Euclidean rotation generator only, and no operator convergence $K_\alpha^W \to K_{\text{cont}}$ is claimed (Observation 5.2).

8.2 Open questions

1. **Other free-field CFTs.** The scalar and Dirac cases exhibit the orthogonal M2/M3 decomposition of Theorem 3.3. Does the same projection structure extend to higher-spin partition functions on S^3, in particular the Maxwell ($p = 1$ form) and graviton ($p = 2$) cases? The Maxwell partition function on round S^3 has known closed forms [25]; testing whether the framework's spectral-zeta machinery reproduces it (and whether the M2/M3 decomposition extends) is a natural follow-up.

2. **Squashed-S^3 generalization.** The KPS values above are for the round (unit) S^3. The squashed-S^3 partition functions of supersymmetric CFTs [24] use deformed half-integer Hurwitz machinery that may not reduce to the framework's MR-B Jacobi-ϑ_2 closed form. Whether the framework can match the squashed case is open.

3. **State-space-GH-rate convergence statement.** The spectral triples converge at the state-space Gromov–Hausdorff level (Paper 38); Observation 5.2 does not inherit an operator-level rate from this (withdrawn 2026-09-28). A quantitative rate for the partition-function-side convergence $|F_{n_{\max}} - F^{\text{KPS}}| \leq C \cdot \gamma_{n_{\max}}$ with explicit C requires a Mellin-domain version of the Berezin-reconstruction framework of Paper 38; this is open work.

4. **Bulk-side identification (Prop. 6.1).** The JLMS bulk-modular Hamiltonian identification, the RT entanglement-entropy formula, and the AdS_4 dual geometry remain open. Importing AdS_4 / H^4 infrastructure is the named major structural extension that would reach these; it is beyond sprint scale.

8.3 Catalogue and open targets

The bit-exact matches at S^3 and S^5 exhibit a pattern: the framework's discrete spectral-zeta machinery reproduces *independently-published* continuum CFT-on-sphere free-energy values, and the transcendental signature decomposes via the master Mellin engine [2]. This pattern admits systematic extension. The following table catalogues the current status:

System	Continuum reference	Fra
Free scalar (conformal) on S^3	KPS [22]	I
Free Dirac (CH) on S^3	KPS [22]	I
Free scalar (conformal) on S^5	Beccaria-Tseytlin [25]	I
Free Dirac (CH) on S^5	Beccaria-Tseytlin [25]	I
Free scalar (conformal) on S^7	— (framework-generated)	DONE (in-ring $\mathcal{R}_7$, $\geq$
Free Maxwell on S^3	3D scalar duality [25]	open: S^3 Maxwell-sc
Free Maxwell on S^5	BT [25]	TODO; needs tran
Free higher-rank tensor ($p \geq 2$) on S^d	BT [25], Anninos	TODO; nee
Free (non-)minimal scalar (zero mode) on S^d	—	open: m=2 edge corre
Squashed S^3 partition function	JKPS, Hama-Hosomichi-Lee	open: deformed half-integer Hu

Holographic Weyl anomaly ↔ Connes–Chamseddine spectral action: both live in M2. A structural classification shared with the AdS/CFT literature that the framework obtains directly — without additional computation — is the M2-sector placement of the holographic Weyl anomaly and the Connes–Chamseddine spectral action.

Henningson and Skenderis [27] derived the Weyl anomaly of CFT_d on a manifold M via the bulk AdS_{d+1} gravitational action regularized at a finite radial cutoff. For even d, the result reproduces the boundary central charges $a_{\mathrm{CFT}}, c_{\mathrm{CFT}}$ as specific combinations of bulk gravitational coefficients. The bulk integral is computed via the Riemannian curvature data of AdS, evaluated through a heat-kernel expansion in the AdS-radius cutoff ϵ.

The Connes–Chamseddine spectral action [29] $\mathrm{Tr}\, f(D/\Lambda)$, on any almost-commutative spectral triple, has heat-kernel expansion coefficients given by Seeley–DeWitt coefficients $a_k(D^2)$. For the GeoVac spectral triple on round S^3, the Seeley–DeWitt coefficients of D^2_{CH} are exact in $\sqrt{\pi}\cdot\mathbb{Q}$ with no higher coefficients (Paper 28 [9], Sprint MR-B closed form, two-term-exact theorem). The spectral action therefore lives in the M2 ring of the master Mellin engine (Paper 18 [2] §III.7).

Structural classification. Both the Henningson–Skenderis holographic Weyl anomaly and the Connes–Chamseddine spectral action sit in the master Mellin engine's M2 sub-mechanism (Seeley–DeWitt heat-kernel ring $\sqrt{\pi}\cdot\mathbb{Q}\oplus\pi^2\cdot\mathbb{Q}$). This is a shared heat-kernel *classification*, not a computed identity: both are heat-kernel-trace observables ($\mathcal{M}[\mathrm{Tr}\, e^{-tD^2}]$ at operator order $k=2$), the HS quantity computed on the bulk-gravity side and the CC quantity on the finite spectral triple. The framework does not compute the HS holographic anomaly; the content is the taxonomic placement of both in the M2 ring. The framework's discrete-cutoff realization (Paper 38 [8] state-space GH convergence at rate $4/\pi$) is the operator-system-level home of the CC side.

This is the framework's most direct contact point with established holographic computations in the M2 sector and is a corollary of the master Mellin engine case-exhaustion theorem (Paper 32 §VIII), not a new computation. The substantive content is the explicit identification of HS's bulk computation as an M2-sector observable, placed in the same heat-kernel ring as the framework's spectral action on S^3 — a structural classification, not a reproduction of the holographic computation.

Relation to the TT-bar cut-off AdS picture. Hartman, Kruthoff, Shaghoulian, and Tajdini [28] showed that sphere partition functions of TT-bar-deformed large-N holographic CFT_d in $d\in\{2,3,4,5,6\}$ non-perturbatively match bulk computations of AdS_{d+1} at finite radial cutoff. The parallel to the present paper's discrete-cutoff framework ($\mathcal{T}_{n_{\max}}$ at finite $n_{\max}$ matching continuum CFT) is suggestive, but the two cut-off structures are *categorically distinct*:

- **TT-bar (Hartman et al. 2019):** a *deformation of the action*. The TT-bar operator $\mathcal{O}_{T\bar{T}}(\mu)$ modifies the field theory's UV at finite μ; the deformed CFT has a different energy spectrum and a different action. The bulk dual is AdS at finite radial cutoff r_c related to μ via a (dimension-dependent) explicit formula.

- **GeoVac $n_{\max}$ (this paper):** a *truncation of the spectrum*, not the action. The framework's discrete spectral triple at $n_{\max}$ keeps the underlying free CFT and its action intact, but restricts the mode sum to $n\le n_{\max}$. As $n_{\max}\to\infty$, the spectral data converges to the round-sphere CFT bit-exactly (Theorems 3.1, 3.2); the intermediate $n_{\max}$ values are not a TT-bar-deformed CFT but genuine spectral truncations of the original CFT.

At the partition-function-value level, the two regularizations might still be related: each produces a finite number Z_{cutoff} that converges to the continuum CFT partition function as the cutoff is removed. Whether there exists a $\mu(n_{\max},R)$ such that $Z^{\mathrm{GeoVac}}_{n_{\max}}(R) = Z_{\text{TT-bar}}(\mu(n_{\max},R),R)$ is an open question. A positive answer would identify the framework's

spectral cutoff with a specific TT-bar-deformation strength; a negative answer would crystallize that the two cutoffs are structurally incompatible at the partition-function level.

Structural distinction from holographic correspondences (dS/CFT and AdS/CFT). The bit-exact reproduction of CFT_3-on-S^3 partition function data (Sec. 3) places the framework in structural adjacency with both AdS/CFT and dS/CFT, but in neither category. Specifically, the framework's S^3 substrate occupies the same geometric position that Euclidean dS/CFT [40] assigns to its boundary CFT_3: in Maldacena's Euclidean sharpening of Strominger's dS/CFT proposal [39], the Hartle–Hawking wavefunctional of dS_4 at I^+ equals the CFT_3-on-S^3 partition function $\Psi_{\mathrm{dS}_4}[\phi_0] = Z_{\mathrm{CFT}_3}[\phi_0]$. The KPS values reproduced bit-exactly here (Theorems 3.1, 3.2) are precisely the boundary CFT data that dS/CFT would predict via this identification.

However, the framework does *not* contain a bulk dS_4 (or its Euclidean partner S^4). The S^3 is the Fock-projection image of the Coulomb spectrum (Paper 7 [1]), not a future-infinity boundary of any 4D bulk; there is no Hartle–Hawking wavefunctional construction (no functional integral over 4D geometries); and there is no holographic dictionary mapping bulk fields to boundary operators. The CFT_3 partition function values emerge directly from spectral asymptotics of the discrete spectral triple, not from holographic reconstruction. This places the framework in a third structural category, distinct from the holographic correspondences:

Theory	Bulk	Correspondence mechanism
$\mathrm{AdS}_5/\mathrm{CFT}_4$	AdS_5 IIB SUGRA	Holographic dictionary
$\mathrm{AdS}_4/\mathrm{CFT}_3$ (ABJM)	AdS_4 M-theory	Holographic dictionary
$\mathrm{dS}_4/\mathrm{CFT}_3$	dS_4	Hartle–Hawking wavefunctional
Chern–Simons / WZW	3D topological	State-sum / topological
Connes–vS (2021) S^1	Discrete spectral triple	Propinquity convergence
GeoVac (this paper)	**Discrete spectral triple on S^3**	**State-space GH convergence at rate $4/\pi$**

The framework's analog of the holographic correspondence is the Paper 38 [8] state-space GH convergence theorem $\Lambda(\mathcal{T}_{n_{\max}}, \mathcal{T}_\infty) \leq C_3 \gamma_{n_{\max}} \to 0$. This is a correspondence between two operator-algebraic objects (the discrete spectral triple at finite cutoff and the continuum spectral triple on round S^3), not between two physical geometries (bulk gravity and boundary CFT). The third category is structurally characterized by:

- the "bulk" replaced by a discrete spectral-triple substrate (operator-system level, finite cutoff);
- the "correspondence" replaced by state-space GH convergence in the continuum limit (Paper 38's unconditional lifted-state proof);
- boundary CFT data emerging from spectral asymptotics (zeta-function derivatives at zero), not from holographic reconstruction.

In one sentence: *the framework's discrete spectral triple reproduces the boundary CFT_3-on-S^3 F-theorem coefficients bit-exactly and yields a wedge-state entropy that counts a lowest-shell degeneracy, while its modular side is an analogy only (the identification of K_α^W as a modular Hamiltonian is open); it is distinguished from AdS/CFT and dS/CFT by absence of any bulk gravitational theory and presence of a state-space-GH-convergence "correspondence" between operator-algebraic discretization levels.* This is not a new holographic correspondence (no bulk = no holography), nor a refutation of dS/CFT (the boundary data is consistent with what dS/CFT would predict); it is a structurally distinct realization that produces the same boundary CFT data via a different mechanism. The bulk-side identifications named in Prop. 6.1 for AdS_4 remain unreachable from the framework's existing infrastructure for the same reasons (no bulk

geometric machinery in `geovac/`). The framework's contribution is to provide a non-holographic operationalization of the boundary-CFT-on-S^3 data, structurally distinct from both AdS/CFT and dS/CFT.

Areas yet to be explored. Beyond the immediate catalogue targets, several adjacent directions extend the framework's reach into established physics:

- **Holographic central charges** (a- and c-anomalies in even d). Henningson-Skenderis [27] and follow-ups give closed forms for the holographic dual of free CFT central charges. The framework's spectral-action machinery (Paper 28 [9]) potentially provides a discrete-cutoff reproduction; the connection to Paper 24's Bargmann-Segal S^5 is unexplored.
- **Entanglement-entropy universal terms.** The continuum Casini-Huerta-Myers [30] entanglement entropy of a ball region has a universal subleading constant equal to $-F$. Connecting this to the framework's wedge KMS structure (Sec. 4) and to the master Mellin engine decomposition would strengthen the F-as-engine-signature reading.
- **Higher-rank Lie groups beyond** SU(2). The Paper 40 unified GH-convergence theorem [10] extends the state-space GH convergence machinery to all simple compact Lie groups. A natural catalogue extension is CFT-on-G partition functions for various G (heat-kernel zeta on SU(3), Sp(2), etc.) where the framework's GH-rate constant $4/\pi$ is universal but the specific transcendental ring may differ.
- **Spectral-action coefficients.** The Connes-Chamseddine [29] spectral action $\mathrm{Tr}\, f(D/\Lambda)$ has heat-kernel expansion coefficients that classify the gravitational + matter content of an almost-commutative spectral triple. These coefficients are exactly the master Mellin engine M2 sector. The catalogue could include M-engine readings of standard spectral-action calculations on S^3, S^5, and higher.
- **Lorentzian-side observables.** The catalogue here is boundary-side Riemannian. The Lorentzian papers (Paper 42; Papers 46–49, now archived or descoped) posited a wedge modular Hamiltonian (its identification remains open) and studied Pythagorean orthogonality and twin-paradox-as-quantum-information structures. Cataloguing Lorentzian-side CFT observables (thermal partition functions on $S^3 \times \mathbb{R}_t$, Rindler-wedge entropies, etc.) is a parallel direction.

Each catalogue entry either (i) verifies the master Mellin engine on independent physics data, (ii) extends a structural cancellation theorem (like log 3 cancellation on S^5, Theorem 7.3), or (iii) identifies a wall where the simple ring fails (like the squashed-S^3 deformed-Hurwitz case, not in the MR-B Jacobi-ϑ_2 closed form). Each outcome is paper-worthy in its own right; the cumulative catalogue would constitute a structural classification of free-field CFT-on-sphere partition functions via the GeoVac spectral triple.

8.4 Place in the GeoVac math.OA series

This paper is the twelfth math.OA standalone in the GeoVac series, following Papers 38, 39, 40, 42, 43 (archived 2026-09-28; its surviving content is Paper 42 § 11), 44, 45, 46, 47, 48, 49. The previous eleven established operator-algebraic legitimacy of the framework's discrete-spectral-triple truncation and the spectral-truncation convergence theory at the Riemannian, tensor-product and universal compact-Lie-group levels, together with the norm-resolvent non-compact-carrier closure at the spectral level. The Lorentzian *propinquity* levels—strong-form Lorentzian (Paper 46) and both synthetic-Lorentzian bridges (Papers 48, 49)—are *descoped* by Paper 45's degeneracy theorem, consistent with this paper's introduction, and are not convergence results;

Papers 46–49 are archived (`docs/retired_papers.md`). The present paper is the *first connecting that math.OA arc to a physics literature* — specifically the CFT-on-sphere partition function literature of [21–23, 25, 26] — with a load-bearing structural finding (the master Mellin engine verification on independent physics data) plus a bit-exact reproduction (partition function), an exact entropy asymptotic (the lowest-shell degeneracy of the posited generator), and one analogy (the posited wedge generator against the Euclidean rotation generator of the CHM construction; the identification with a modular Hamiltonian is open).

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